NFL BIG DATA BOWL 2026 / ANALYTICS REPORT

THE HIDDEN SPACE

Testing Tyreek Hill's Defensive Gravity

Does defensive attention around Tyreek Hill correspond to more space for Travis Kelce, and does that relationship change under man and zone coverage?

203	3	4
final matched plays	tracking metrics	analysis outputs

Executive summary

We expected Kelce to have more room when multiple defenders were close to Hill. Our data did not show a reliable increase. Kelce averaged 2.64 yards of separation under high Hill attention and 3.24 yards under low attention, but the gap was largely related to the mix of man and zone coverage in those groups.

THE FINDING TO REMEMBER

Kelce had considerably more space against zone coverage (3.58 yd on average) than man coverage (1.78 yd), while the Hill-attention difference within each scheme was small and uncertain. This is an association, not evidence that coverage type caused the difference.

What this report contains

Three measurement methods, four result charts, football takeaways for coaches, scouts and broadcasters, plus limitations and links to the reproducible code.

Study scope: 2021 Kansas City Chiefs passing plays, Weeks 1–8. Player tracking and PFF scouting labels. Analysis branch: Data+Metrics.

01 / RESEARCH SETUP

Data and measurement

One consistent comparison point: 1.5 seconds after the snap.

The tracking data records player positions 10 times per second. We analyzed Kansas City plays in which both Tyreek Hill and Travis Kelce ran routes. The coordinates were already normalized so that the offense moves in the same direction on each play.

Dataset stage	Eligible plays	How it is used
Metric 1: attention	236	Hill and coverage defenders measured at 1.5 seconds
Metric 2: separation gain	204	Pass plays with snap-to-1.5-second measurements
Metric 3: gravity relationship	236	Kelce separation compared with Hill attention
Final matched sample	203	Plays present in all three files; the four outputs use this set

One Metric 2 play had a pass thrown before 1.5 seconds, so it could not be matched to a valid pre-throw 1.5-second snapshot. Other plays were excluded because they lacked a qualifying thrown pass or eligible measurement.

The three core metrics

Metric	Our definition	Important caveat
1. Defensive Attention	Unique coverage defenders within 5 yards of Hill at 1.5 seconds. High = 2+; low = 0–1.	Distance is not proof of a double-team.
2. Space Exploitation	Kelce distance to nearest coverage defender; gain = 1.5-second separation minus snap separation.	More space does not guarantee a safe throwing lane.
3. Gravity Relationship	Compare Kelce separation when Hill attention is high versus low.	Association is not causation.

Quality checks

We matched plays using gameId and playId, checked for duplicate records and missing values, compared coverage labels, and confirmed that the two 1.5-second separation values agreed within rounding. Where shown, uncertainty intervals use 10,000 bootstrap resamples.

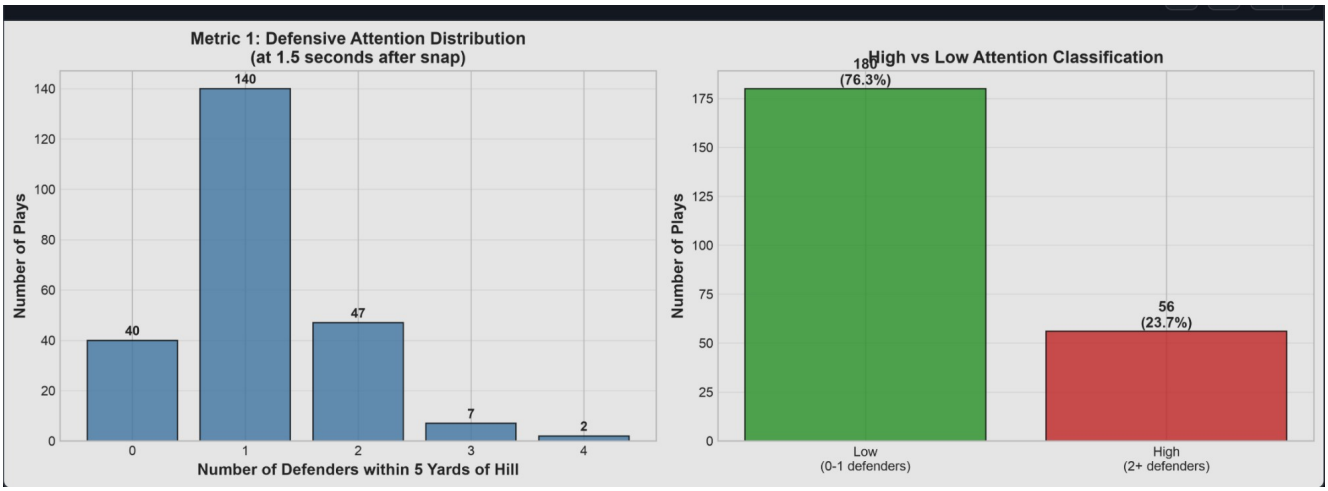
HOW TO READ THE CHARTS

All four outputs use the same 203 plays. The standalone metric charts may use larger source samples. A receiver is marked "open" when his nearest coverage defender is at least 3 yards away; this is a practical cutoff, not an official NFL completion probability.

02 / METRIC 1

Defensive Attention

How many coverage defenders were within five yards of Tyreek Hill?



Metric 1 (236 plays). Defender count at 1.5 seconds after the snap, plus high versus low classification.

Defenders near Hill	Plays	Share
0	40	16.9%
1	140	59.3%
2	47	19.9%
3	7	3.0%
4	2	0.8%

Across Metric 1's 236 plays, Hill had 0–1 nearby coverage defenders on 180 plays (76.3%) and 2+ on 56 plays (23.7%). In the 203-play final shared set, the groups are 158 low-attention and 45 high-attention plays.

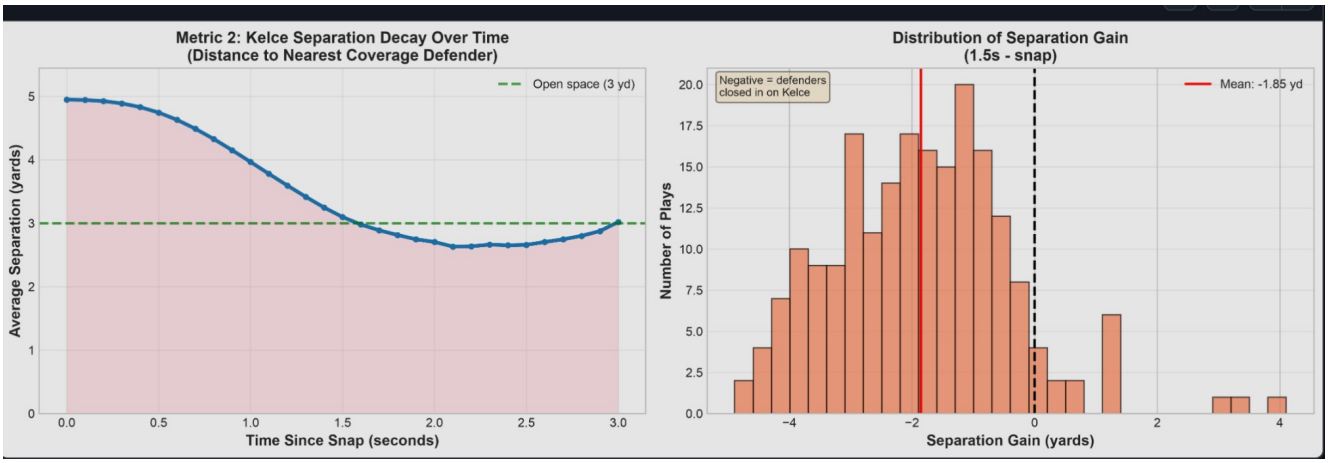
WHAT WE LEARN

Multiple coverage defenders are not usually within five yards of Hill at the measurement time. This describes positions only; it does not show who was actually assigned to cover him.

03 / METRIC 2

Space Exploitation

How quickly does the room around Travis Kelce change?



Metric 2. Kelce's mean nearest-defender distance over time and the distribution of snap-to-1.5-second changes.

Measurement (203 final plays)	Result
Average separation at snap	4.95 yd
Average separation at 1.5 seconds	3.11 yd
Average change	−1.84 yd
Plays with positive separation gain	15 / 203 (7.4%)

Kelce usually began with more room and had less space 1.5 seconds later. This does not mean he was running routes poorly; defenders can start several yards away and close in naturally once the play develops.

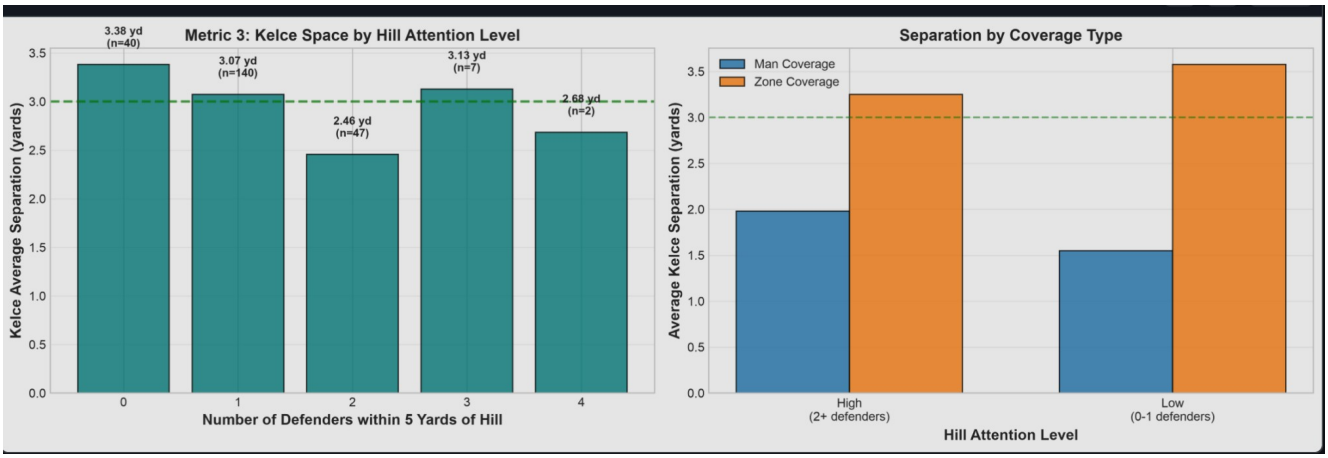
CAREFUL INTERPRETATION

The frame-by-frame curve can include a changing mix of plays at later times, since shorter plays finish sooner. Our headline numbers use the consistent snap-to-1.5-second comparison for eligible plays.

04 / METRIC 3

Defensive Gravity Relationship

Does Hill being crowded line up with more space for Kelce?



Metric 3 (236 plays). Kelce separation by Hill's nearby defender count, and a man/zone split.

Final 203-play comparison	Kelce separation
High Hill attention (45 plays)	2.64 yd
Low Hill attention (158 plays)	3.24 yd
Difference: high minus low	−0.60 yd
Spearman correlation	$\rho \approx -0.08$ ($p \approx 0.27$)

The overall relationship is very weak. The result does not support a reliable claim that drawing more nearby defenders toward Hill gives Kelce more room. The coverage breakdown is essential, because man and zone plays are not distributed evenly between the attention groups.

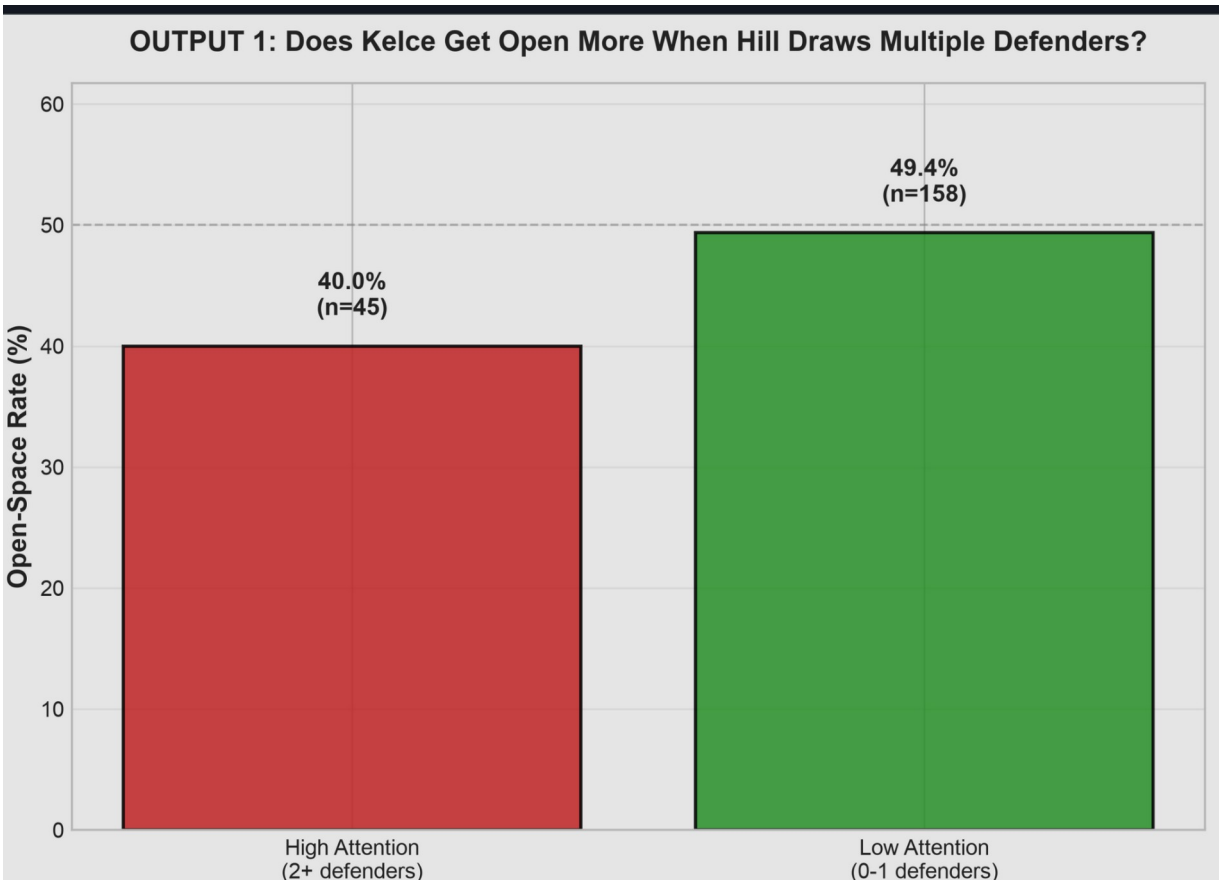
WHAT CHANGES THE STORY

When we separate man and zone, the Hill-attention differences become smaller and even point in opposite directions. We therefore need to report the coverage split, not just the overall −0.60-yard gap.

05 / OUTPUT 1

Open-Space Rate

Does Kelce have at least three yards of space more often when Hill has multiple nearby defenders?



Output 1 (203 plays). Percentage meeting the 3-yard threshold under high versus low Hill attention.

Hill attention	Open plays	Rate
High (2+ defenders)	18 / 45	40.0%
Low (0–1 defenders)	78 / 158	49.4%

The rate was 9.4 percentage points lower with high Hill attention. However, the 95% bootstrap interval for the difference ranged from approximately −25.5 to +7.1 percentage points, and Fisher's exact test gave $p \approx 0.31$.

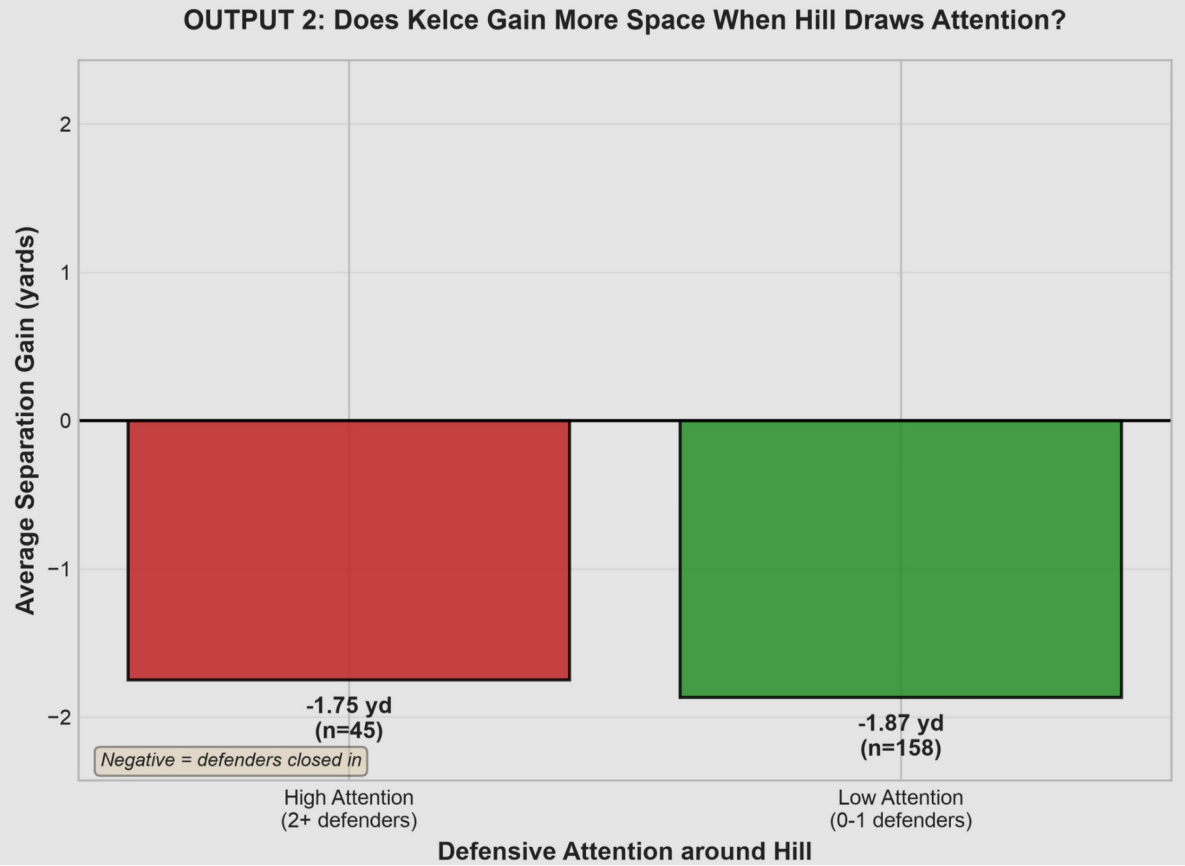
FINDING

The observed rate is lower, but the evidence is not strong enough to say the difference is reliable.

06 / OUTPUT 2

Average Separation Gain

Does Kelce gain more space in the first 1.5 seconds when Hill attracts additional nearby defenders?



Output 2 (203 plays). Negative values mean Kelce's nearest defender was closer at 1.5 seconds than at the snap.

Hill attention	Mean separation gain	Positive-gain plays
High (45 plays)	−1.75 yd	1 / 45 (2.2%)
Low (158 plays)	−1.87 yd	14 / 158 (8.9%)

Both groups lost separation. The average difference between groups was only +0.12 yards. Its 95% bootstrap interval was about −0.28 to +0.50 yards (Mann–Whitney $p \approx 0.28$).

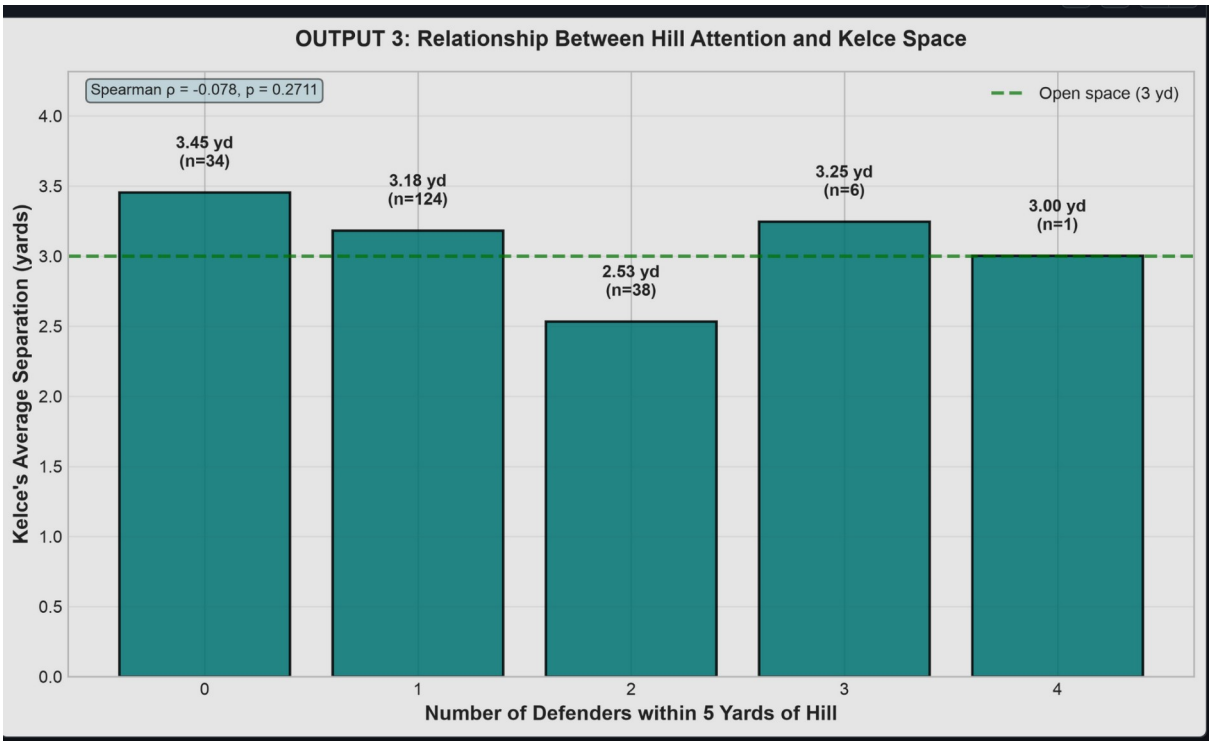
FINDING

Defenders closed in on Kelce in both conditions. There is no convincing difference in separation gain based on Hill's nearby defender count.

07 / OUTPUT 3

Attention and Available Space

Does Kelce have more room as the number of defenders near Hill increases?



Output 3 (203 plays). Mean Kelce separation against the count of nearby defenders around Hill.

Nearby defenders	Plays	Average Kelce separation
0	34	3.45 yd
1	124	3.18 yd
2	38	2.53 yd
3 or more	7	3.21 yd

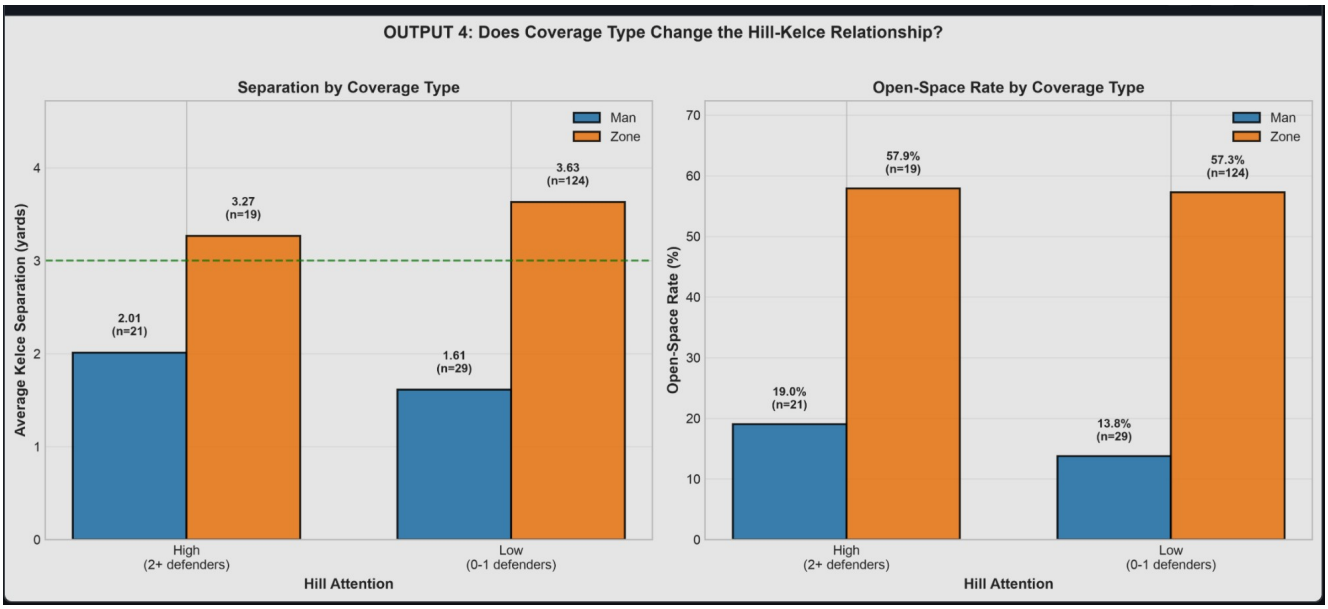
There is no smooth upward trend: Spearman $\rho \approx -0.08$, $p \approx 0.27$. The final group combines 3 and 4 defenders because only seven plays fell into that range.

The high-versus-low mean difference was -0.60 yards, with a percentile bootstrap interval around -1.18 to -0.01 yards; a Mann–Whitney comparison gave $p \approx 0.13$. Because these methods assess different properties of the data, and some plays share a game, we treat the finding as uncertain rather than claiming a definitive effect.

08 / OUTPUT 4

Man vs Zone Coverage

The clearest pattern in our analysis.



Output 4. Kelce separation (left) and 3-yard open-space rates (right), split by scheme and Hill attention.

Coverage	Hill attention	Plays	Sep.	Open rate
Man	High	21	2.01 yd	19.0%
Man	Low	29	1.61 yd	13.8%
Zone	High	19	3.27 yd	57.9%
Zone	Low	124	3.63 yd	57.3%

Ten "Other" coverage plays were left out of this chart. Within man coverage, high Hill attention was linked to +0.40 yd ($p \approx 0.34$); within zone, it was linked to −0.36 yd ($p \approx 0.75$). Neither within-scheme comparison was conclusive.

Across all man and zone plays in the shared sample, Kelce had about 1.78 yards of separation against man and 3.58 yards against zone. High-attention plays were much more often man coverage (21 of 40 man/zone high-attention plays) than low-attention plays (29 of 153), helping explain why the overall high-attention average looked worse.

09 / FOOTBALL RELEVANCE

What this means on the field

What coaches, scouts and broadcasters can take away - without overstating the evidence.

For coaches

The count of defenders near a star receiver is not enough to tell you whether another player has room. Against these Chiefs plays, Kelce generally had more separation when the defense played zone. Coaches could use this type of tracking view to study which routes find pockets of space against specific coverage schemes. Our results do not prove that a coach can force zone coverage.

For scouts

Traditional statistics tell us who caught the ball; tracking can also describe where players find room. But a proper player grade would need more than nearest-defender distance. It would also need route type, assignment, timing, starting alignment and whether a pass was realistically available.

For broadcasters

“We thought two defenders near Tyreek Hill might give Travis Kelce more room. But in these Chiefs plays, Kelce tended to have more space against zone coverage, regardless of how many defenders were near Hill.”

Limitations worth telling the judges

- A defender within five yards of Hill may be covering an area rather than Hill himself.
- Three yards from the nearest defender does not guarantee a clean throwing lane or a successful pass.
- Coverage type, routes, down and distance, field position and pass timing can all shape the result.
- Some subgroups are small (for example, 19 zone/high plays and seven plays with 3+ nearby defenders).
- The study covers one team for the first eight weeks of 2021 and uses a selected subset of plays with valid pre-throw tracking.
- Plays within the same games share context, so ordinary resampling can understate uncertainty.

FINAL CONCLUSION

Our analysis did not clearly support the original defensive-gravity hypothesis. The more useful finding is that Kelce had considerably different amounts of space against man and zone coverage. The number of defenders near Hill, on its own, was not a reliable shortcut for Kelce's opportunity.

10 / REPRODUCIBILITY

Methods, code and sources

Where the measurements and charts came from.

Analysis workflow

- Filtered 2021 Chiefs plays where both Hill and Kelce ran pass routes.
- Measured positions from normalized x/y coordinates 1.5 seconds after the snap.
- Counted opposing coverage defenders near Hill and measured Kelce-to-nearest-defender distance.
- Matched the three datasets on gameId and playId; checked missing values, uniqueness, coverage types and rounded separation agreement.
- Compared overall results and man/zone groups; reported sample sizes and uncertainty where available.

Repository files

Component	Path
Defensive attention	src/metric1_defensive_attention.py
Space exploitation	src/metric2_kelce_separation.py
Gravity relationship	src/metric3_gravity_space_relationship.py
Final four charts	src/final_outputs_analysis.py
Final sample	data/processed/final_analysis.csv
Metric CSVs	data/processed/metric1_attention.csv, metric2_separation_gain.csv, metric3_gravity_space.csv
Charts	outputs/*.png

Source and project link

GitHub: https://github.com/rxd5484/nfl-big-data-ars-2026 (analysis branch: Data+Metrics)

Tracking source: 2021 NFL Big Data Bowl regional-event tracking dataset (Weeks 1–8), with PFF role and coverage labels. Source repository: https://github.com/ThompsonJamesBliss/nfl-big-data-bowl-regional-event-data/

This report uses the team's existing analysis and chart screenshots, including the seven supplied visuals. All reported findings are observational; they should not be interpreted as proof that Hill or the coverage scheme caused Kelce's separation.